

## MATHEMATICAL OPTIMIZATION: QUESTION LIST

ABSTRACT. Some notes on organization and a list of questions helping you to prepare for the exam. At the exam, write formulas, write definitions, explain concepts.

### COURSE ORGANIZATION

- 2h lectures + 2h Lab exercises per week.
- Bring your laptop to the lab.
- 80% lab attendance to pass the exercises.
- Office hours: Monday 2-3pm, Room 4.19
- [felix.huber@ug.edu.pl](mailto:felix.huber@ug.edu.pl) to contact me by email.  
Write class (MathOpt) and surname in subject.
- Exam: 2h written exam; at most 1 retake. Cheating (or trying to do so) results in a fail.

### MATERIAL

- (1) Lovász, lecture notes: Semidefinite programs and combinatorial optimization  
[https://sites.math.washington.edu/~thomas/teaching/m514\\_web/Lovasz\\_semidef.pdf](https://sites.math.washington.edu/~thomas/teaching/m514_web/Lovasz_semidef.pdf)
- (2) Gupta, O'Donnell - CMU's course on LP and SDP (our convention)  
<https://www.cs.cmu.edu/afs/cs.cmu.edu/academic/class/15859-f11/www/notes/lpsdp.pdf>
- (3) Matoušek, Gärtner - Understanding and using Linear Programming <https://link.springer.com/content/pdf/10.1007/978-3-540-30717-4.pdf>
- (4) Heng Yang - Semidefinite Optimization and Relaxation <https://hankyang.seas.harvard.edu/Semidefinite/> (careful: different convention for primal and dual)
- (5) Blekherman et al, Semidefinite Optimization and convex algebraic geometry  
<https://www.mit.edu/~parrilo/sdocag/>
- (6) JuMP library for Optimization <https://jump.dev/>

**For all questions use notation/definitions from our class.**